HW 4 Solution

Written Problems (total 60 points)

1.(15 points)

2.1.6

Consider the problem

$$\text{maximize} \quad x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$$

$$\text{subject to} \quad \sum_{i=1}^{n} x_i = 1, \qquad x_i \geq 0, \quad i = 1, \ldots, n,$$

where a_i are given positive scalars. Find a global maximum and show that it is unique.

Solution:

optimality condition. (5 points)

find a global maximum. (5 points)

show it is unique. (5 points)

2.1.6

Solution: maximize the function $x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ is equal to maximze $\ln(x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n})$, which is also equal to minimize the function $-\ln(x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}) = -\sum_{i=1}^{n} a_i \ln x_i$

Let $f(x) = -\sum_{i=1}^{n} a_i \ln x_i$, s.t. $\sum_{i=1}^{n} x_i = 1$, $x_i \geq 0$

Since $\nabla^2 f(x) = \begin{bmatrix} \frac{a_1}{x_1^2} & & \\ & \frac{a_2}{x_2^2} & \\ & & \ddots & \\ & & & \frac{a_n}{x_n^2} \end{bmatrix} \succeq 0$, thus $f(x)$ is a convex function

Also, we have $\frac{\partial f(x)}{\partial x_i} = -\frac{a_i}{x_i}$

And we know that when $x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ is maximized, $x_i > 0$ must hold for all i, thus for the optimal solution x^*, we have $x_i^* > 0$

Also, we know that for $x_i^* > 0 \Rightarrow \frac{\partial f(x^*)}{\partial x_i} \leq \frac{\partial f(x^*)}{\partial x_j} \; \forall j$

Since $x_i^* > 0$ for all i, thus we also have $\frac{\partial f(x^*)}{\partial x_i} \geq \frac{\partial f(x^*)}{\partial x_j} \; \forall j$

Thus $\frac{\partial f(x^*)}{\partial x_i} = \frac{\partial f(x^*)}{\partial x_j} \; \forall i, j \Rightarrow -\frac{a_i}{x_i^*} = -\frac{a_j}{x_j^*} \; \forall 1 \leq i, j \leq n$

Since $\sum_{i=1}^{n} x_i = 1$ and $\frac{a_i}{x_i^*} = \frac{a_j}{x_j^*} \; \forall i, j$. we have $x_i^* = \frac{a_i}{\sum_{k=1}^{n} a_k}$ $i = 1, 2 \cdots n$ is the optimal solution, and also it's the only optimum, since $\nabla^2 f(x)$ is strictly positive definite for all $x > 0$, the optimum is unique.

2.(20 points)

2.1.12 (Projection on a Simplex)

(a) Develop an algorithm to find the projection of a vector z on a simplex. This algorithm should be almost as simple as a closed form solution.

Solution:
optimality condition. (5 points)
algorithm. (10 points)
as simple as a closed form solution. (5 points)

2.1.12

(a)

$$\min_x \underbrace{\frac{1}{2}\|x-z\|^2}_{f(x)}$$

s.t. $x \in X = \{x \mid x \geq 0, \sum_{i=1}^{r} x_i = r\}$

By the necessary condition for optimization over simplex, see (2.10)

$$x_i^* > 0 \Rightarrow \frac{\partial f(x^*)}{\partial x_i} \leq \frac{\partial f(x^*)}{\partial x_j}, \ \forall j$$

Suppose $x_i^* > 0$, $x_j^* = 0$

$$\Rightarrow \frac{\partial f(x^*)}{\partial x_i} = x_i^* - z_i \leq x_j^* - z_j = -z_j = \frac{\partial f(x^*)}{\partial x_j}$$

$$\Rightarrow z_i > z_j + x_i^* \geq z_j \quad (\text{since } x_i^* \geq 0, \forall i)$$

WLOG, assume $z_1 \geq z_2 \geq \cdots \geq z_n$

and $x_1^*, x_2^*, \ldots, x_m^* > 0$, $x_{m+1}^*, x_{m+2}^*, \ldots, x_n^* = 0$, $1 \leq m \leq n$

$$\Rightarrow \exists \text{ a costant } \lambda = \frac{\partial f(x^*)}{\partial x_i} = x_i^* - z_i, \quad i = 1, \cdots, m$$

$$\Rightarrow \lambda = x_1^* - z_1 = x_2^* - z_2 = \cdots = x_m^* - z_m$$

summation over $(x_i^* - z_i) \Rightarrow \sum_{i=1}^{m}(x_i^* - z_i) = \sum_{i=1}^{n} x_i^* - \sum_{i=1}^{m} z_i = m \cdot \lambda \Rightarrow \lambda = \dfrac{r - \sum_{i=1}^{m} z_i}{m}$

Thus, $x_j^* = z_j + \lambda = z_j + \dfrac{r - \sum_{i=1}^{m} z_i}{m} > 0$, $j = 1, \cdots, m$

Steps:

1. sort $\underline{z}$ as $z_1 \geq z_2 \geq \cdots \geq z_n$
2. $m = \arg\max j$ s.t. $z_j + \dfrac{r - \sum_{i=1}^{j} z_i}{j} > 0$
3. $x_i^* = z_i + \dfrac{r - \sum_{i=1}^{m} z_i}{m}$, $i = 1, \cdots, m$

   $x_i^* = 0$, $i = m+1, \cdots, n$

3.(10 points)

2.3.1 (Scaled Gradient Projection Method)

Let H^k be a positive definite matrix and define $\bar{x}^k$ by

$$\bar{x}^k = \arg\min_{x\in X} \left\{ \nabla f(x^k)'(x-x^k) + \frac{1}{2s^k}(x-x^k)'H^k(x-x^k) \right\}.$$

Show that $\bar{x}^k$ solves the problem

$$\begin{aligned} &\text{minimize} \;\; \tfrac{1}{2}\left\| x - \left(x^k - s^k (H^k)^{-1}\nabla f(x^k)\right)\right\|_{H^k}^2 \\ &\text{subject to} \;\; x \in X, \end{aligned}$$

where $\|\cdot\|_{H^k}$ is the norm defined by $\|z\|_{H^k} = \sqrt{z'H^k z}$.

Solution: (10 points)

Let

$$F_k(x) = \nabla f(x^k)'(x-x^k) + \frac{1}{2s^k}(x-x^k)'H^k(x-x^k),$$
$$G_k(x) = (x - x^k + s^k(H^k)^{-1}\nabla f(x^k))'H^k(x - x^k + s^k(H^k)^{-1}\nabla f(x^k)).$$

Then

$$\begin{aligned} G_k(x) &= (x-x^k)'H^k(x-x^k) + 2s^k\nabla f(x^k)'(x-x^k) + (s^k)^2\nabla f(x^k)'(H^k)^{-1}\nabla f(x^k) \\ &= 2s^kF_k(x) + (s^k)^2\nabla f(x^k)'(H^k)^{-1}\nabla f(x^k). \end{aligned}$$

Since s^k and $(s^k)^2\nabla f(x^k)'(H^k)^{-1}\nabla f(x^k)$ are constant,

$$\min_{x\in X} F_k(x) \Longleftrightarrow \min_{x\in X} G_k(x).$$

That is, $\bar{x}^k$ minimizes $F_k(x)$ over X if and only if $\bar{x}^k$ minimizes $G_k(x)$ over X. This completes the proof.

4.(15 points)

3.1.1

Use the Lagrange multiplier theorem to solve the following problems:

(a) $f(x) = ||x||^2$, $h(x) = \sum_{i=1}^{n} x_i - 1$.

(b) $f(x) = \sum_{i=1}^{n} x_i$, $h(x) = ||x||^2 - 1$.

(c) $f(x) = ||x||^2$, $h(x) = x'Qx - 1$, where Q is positive definite.

Solution: (5 points each)

Solution:

(a) We have the Lagrangian Multiplier as

$L(x,\lambda) = f(x) + \lambda h(x)$. for the local min, we should have

$\nabla L(x,\lambda) = 0 \Rightarrow \frac{\partial f(x)}{\partial x_i} + \lambda \frac{\partial h(x)}{\partial x_i} = 2x_i + \lambda = 0$, $\sum_{i=1}^{n} x_i - 1 = 0$

Then we have the unique solution as $x_i^* = \frac{1}{n}$, ~~and the~~ $i = 1, 2 \cdots n$

and the optimal value is $f(x^*) = \frac{1}{n}$

(b) We have the Lagrangian Multiplier as:

$L(x,\lambda) = f(x) + \lambda h(x)$, for local min we should have

$\frac{\partial f(x)}{\partial x_i} + \lambda \frac{\partial h(x)}{\partial x_i} = 1 + 2\lambda x_i = 0$, $\sum_{i=1}^{n} x_i^2 - 1 = 0$

Then we have two solutions $x_i = \frac{1}{\sqrt{n}}$, $\lambda = -\frac{\sqrt{n}}{2}$ and $x_i = -\frac{1}{\sqrt{n}}$, $\lambda = \frac{\sqrt{n}}{2}$

By comparison, we know the optimum should be $x_i = -\frac{1}{\sqrt{n}}$ $i = 1, 2 \cdots n$

(c) We have the Lagrangian Multiplier as:

$L(x,\lambda) = f(x) + \lambda h(x) = ||x||^2 + \lambda(x^T Q x - 1)$, for local min we should have

$\nabla L(x,\lambda) = 0 \Rightarrow \nabla f(x) + \lambda \nabla h(x) = 0$, $h(x) = 0$

Then we have $2x + 2\lambda Q x = 0$, $x^T Q x - 1 = 0 \Rightarrow -\frac{1}{\lambda} x = Qx$

Multiply the first term by x^T, we have $2x^T x + 2\lambda x^T Q x = 0 \Rightarrow ||x||^2 = -\lambda$

Thus, x is eigen vector for the ~~matrix~~ eigen value of matrix Q that has the Largest absolute value, and $||x||^2 = -\lambda$, where $-\frac{1}{\lambda}$ is the eigen value